Week 6 guest lecture: Diffusion models for NLP

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27 February 2026

- Revisiting autoregressive language models
  - A latent variable view
  - Generation order and masked language modelling
- Very brief introduction to diffusion models
- Three approaches to diffusion for language
- Practical considerations and the state of the art
  - Beyond unconditional text generation
- Remarks and conclusion

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Autoregressive language models

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$$p(\text{The cats are asleep}) =$$
$$= p(\text{The} \mid \ll\gg) p(\text{cats} \mid \text{The}) p(\text{are} \mid \text{The cats}) p(\text{asleep} \mid \text{The cats are})$$

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- How many model passes to generate N words? $O(N)$
  - n-gram model, RNN (past states cached): $O(1)$ for k-th word $\rightsquigarrow$ $O(N)$ operations
  - Transformer: $O(k^2)$ computation for k-th word $\rightsquigarrow$ $O(N^3)$ operations

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Motivation I: Can we generate text in fewer that $O(\text{length})$ model passes?

Some weaknesses of autoregressive generation

- **Recency bias:** long-range dependencies hard to model

A: I'm Natasha. I study neural language models and dialog systems. Are you an AI researcher too?
B: No, though I do like chatting with bots and laughing at their mistakes. But what was your name again?
A: Oh, you forgot already? My name is [w]

$p(w \mid \text{context})$ *1.* Alex (1.9%) *2.* **Natasha** (1.7%) *3.* also (1.5%)

Ballad metre is "less regular and more conversational" than common [w]

$p(w \mid \text{context})$ *1.* sense (9.0%) *2.* in (2.0%) *3.* . (1.9%) . . . *13.* **metre** (0.6%)

Isley Brewing Company: Going Mintal – a minty milk chocolate [w]

$p(w \mid \text{context})$ *1.* bar (4.8%) *2.* drink (3.7%) *3.* with (3.5%) . . . *13.* **stout** (2.7%)

Other times anxiety is not as easy to see, but can still be just as [w]

$p(w \mid \text{context})$ *1.* important (5.6%) *2.* bad (4.6%) *3.* **debilitating** (4.3%)

[Malkin et al., 'Coherence. . . ', ACL'22]
see also [Khandelwal et al., 'Sharp nearby. . . ', ACL'18]

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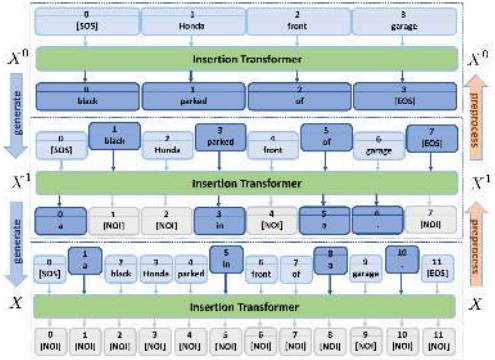

[Zhang et al., 'POINTER...', EMNLP'20]

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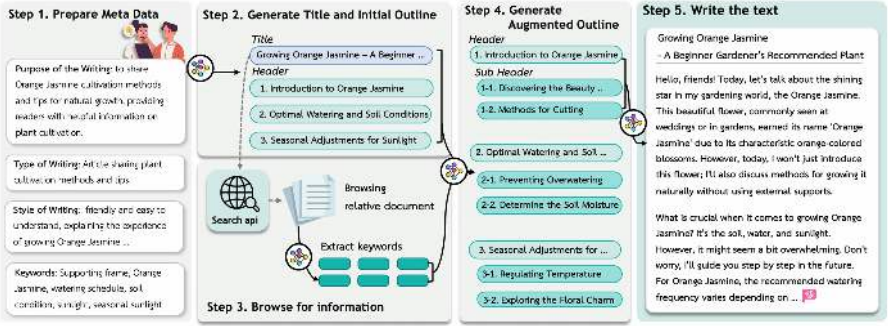

[Lee et al., 'Navigating the path. . . ', NAACL'25]

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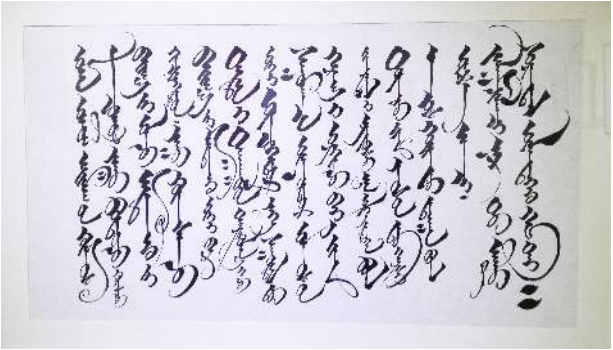
Motivation II: Can we generate text in not necessarily left-to-right* order?

A small digression: ‘left to right’ is not universal



몽골 문자(몽골어: 몽골 비치그, монгол бичиг)는 몽골어를 위해 만들어진 여러 표기 체계들 가운데 하나이다. 약간만 변경된 채로, 이것은 몽골어와 에벤크어를 적기 위해 오늘날까지 중화인민공화국의 내몽골 자치구에서 사용되고 있다.

More inclusive: ‘streaming / temporal / causal / logical / phonetic order’

A latent variable view

Instead of next-word probabilities...

$$p(\text{The} \mid \ll\gg)p(\text{cats} \mid \text{The})p(\text{are} \mid \text{The cats})p(\text{asleep} \mid \text{The cats are})$$

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Think of state transitions:

$$\overbrace{\ll\gg}^{z_4} \rightsquigarrow \overbrace{\text{The}}^{z_3} \rightsquigarrow \overbrace{\text{The cats}}^{z_2} \rightsquigarrow \overbrace{\text{The cats are}}^{z_1} \rightsquigarrow \overbrace{\text{The cats are asleep}}^{z_0}$$

$$p(z_0) = p(z_4)p(z_3 \mid z_4)p(z_2 \mid z_3)p(z_1 \mid z_2)p(z_0 \mid z_1)$$

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Or use masked states:

$$\overbrace{\boxed{\text{M}}\boxed{\text{M}}\boxed{\text{M}}\boxed{\text{M}}}^{z_4} \rightsquigarrow \overbrace{\text{The } \boxed{\text{M}}\boxed{\text{M}}\boxed{\text{M}}}^{z_3} \rightsquigarrow \overbrace{\text{The cats } \boxed{\text{M}}\boxed{\text{M}}}^{z_2} \rightsquigarrow \overbrace{\text{The cats are } \boxed{\text{M}}}^{z_1} \rightsquigarrow \overbrace{\text{The cats are asleep}}^{z_0}$$

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Only z_0 is observed; $z_1, \ldots, z_4$ are latent (hidden) states

The problem of generation order

Why generate from left to right (and not in some other order)?

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Why generate from left to right (and not in some other order)?

- We speak left to right; interested in open-ended generation
- Word order is a convention for ease of production and processing
  - Related work on linguistic universals: much of syntax can be explained by information-theoretic principles
    e.g., [Hahn, Jurafsky, & Futrell, PNAS, 2020]
  - Computational evidence in support
    e.g., [Pham et al., 'Out of order. . . ', ACL'21] [Sinha et al., 'Masked language modeling. . . ', EMNLP'21] [Malkin et al., 'Studying word order. . . ', EMNLP'21]
- Left-to-right language models have lower perplexity

Generating text in any order with masked LMs

- Train a model to predict all missing words given an arbitrarily masked text
  - Like BERT with all possible masks

z_2	M	cats	M	asleep
top $p(w_i \mid z_2)$	The		are	
	My		fell	
	Some		fall	
	Are		not	

z_3	M	M	are	M
top $p(w_i \mid z_3)$	The	many		there
	How	much		you
	Some	old		they
	My	cats		big

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- Unmask words in a random order (one per step)

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$$\rightsquigarrow \text{The cats}\ \boxed{\text{M}}\ \text{asleep} \qquad p(z_1 \mid z_2)$$

$$\rightsquigarrow \text{The cats are asleep} \qquad p(z_0 \mid z_1)$$

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This is a form of a diffusion model (remember this slide)

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Diffusion models: Overview

- A kind of continuous-time generative model, state-of-the-art in image generation and other domains
- The classical references (mainly for image data):
  - [Sohl-Dickstein et al., 'Deep unsupervised learning. . . ', ICML'15]
  - [Ho et al., 'Denoising diffusion. . . ', NeurIPS'20]
  - [Song et al., 'Score-based generative modeling. . . ', NeurIPS'21]
  - [Dhariwal & Nichol, 'Diffusion models beat GANs. . . ', NeurIPS'21]
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  - Mini-course on the history and latest developments (ask for slides and videos)
- Main properties:
  - Generation by iterative resampling of a state representing an entire object
  - Adaptive (but long) inference time
  - Efficient and stable training
  - No exact likelihood computation

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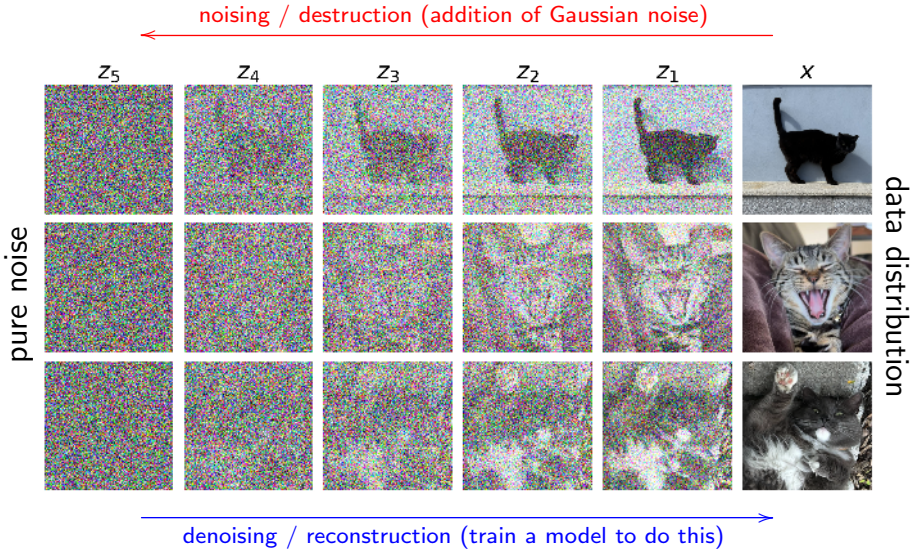

Diffusion models: Training and sampling

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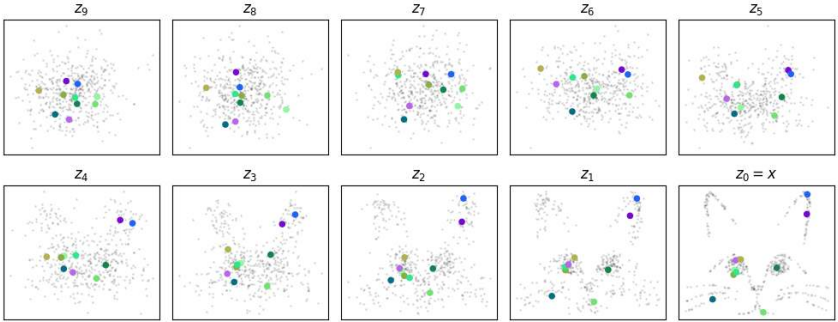

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  - Gradient step on its log-likelihood under p_θ:

$$\ell(x; \theta) = \mathbb{E}_{z_1,\ldots,z_n} \log \left[p_\theta(z_N) p_\theta(z_{N-1} \mid z_N) \ldots p(x \mid z_1)\right]$$

  Magic property: this simplifies to recovering x from a single z_n, for each n
  - After training, we can sample from p_θ in reverse: $z_N \rightsquigarrow z_{N-1} \rightsquigarrow \cdots \rightsquigarrow x$

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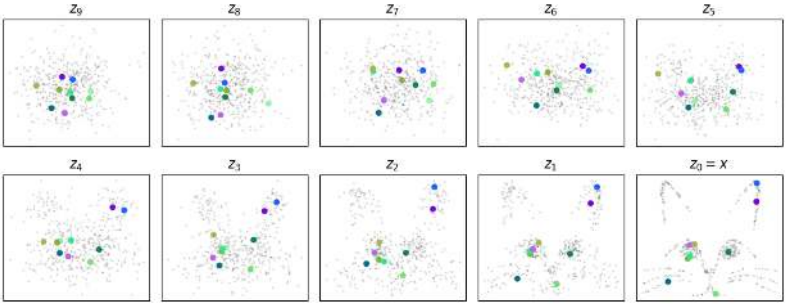
z_N \arrow[r, "p_\theta(z_{N-1}|z_N)"] & z_{N-1} \arrow[l, bend left, dashed, "q(z_N|z_{N-1})"] \arrow[r, "p_\theta(z_{N-2}|z_{N-1})"] & \cdots \arrow[l, bend left, dashed, "q(z_{N-1}|z_{N-2})"] \arrow[r] & z_1 \arrow[r, "p_\theta(x|z_1)"] & z_0 = x \arrow[l, bend left, dashed, "q(z_1|x)"]
\end{tikzcd}$$

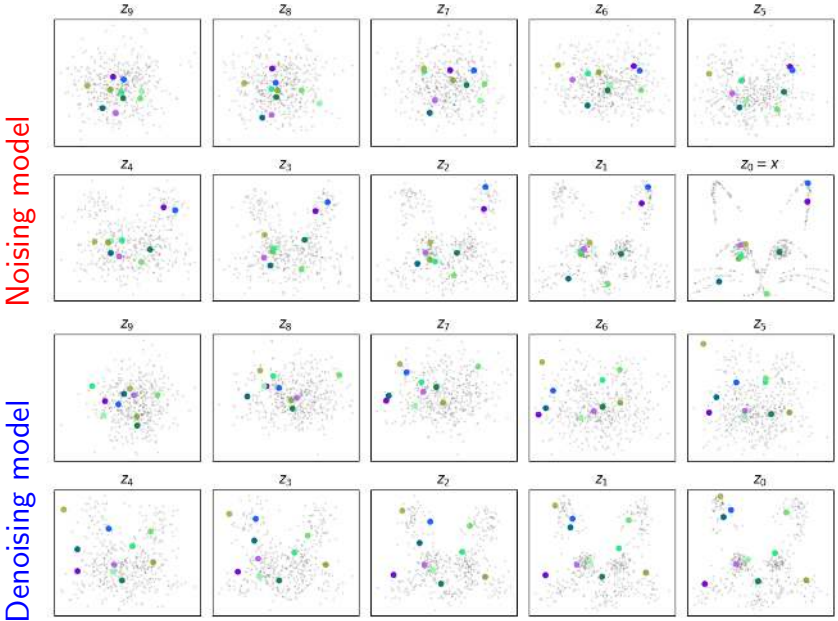
- Diffusion models (typically) fix the distributions $q(z_n \mid z_{n-1})$
  - Typically to adding Gaussian noise: $q(z_n \mid z_{n-1}) = \mathcal{N}(z_n; z_{n-1}, \sigma_n^2 I_d)$ or similar
- Training:
  - Sample the noising trajectory using q: $x = z_0 \rightsquigarrow z_1 \rightsquigarrow \cdots \rightsquigarrow z_N$
  - Gradient step on its log-likelihood under p_θ:

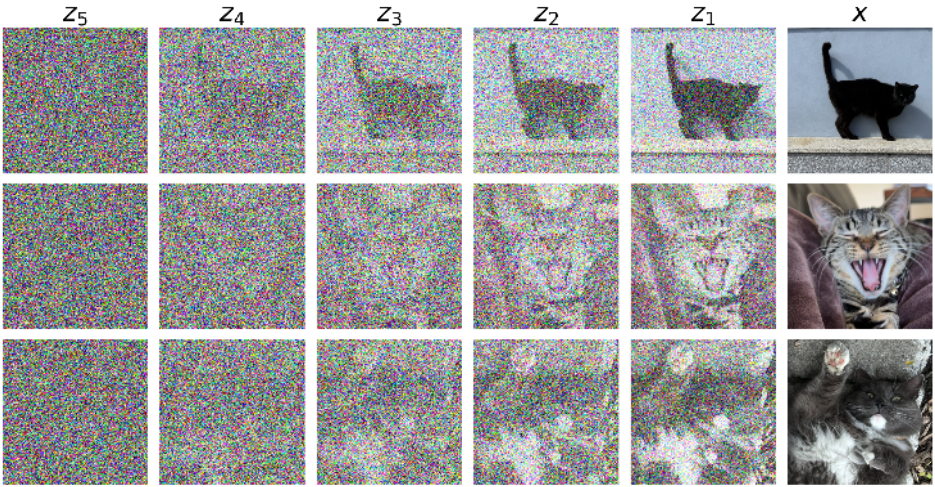
$$\ell(x; \theta) = \mathbb{E}_{z_1,\ldots,z_n} \log \left[p_\theta(z_N) p_\theta(z_{N-1} \mid z_N) \ldots p(x \mid z_1)\right]$$

    Magic property: this simplifies to recovering x from a single z_n, for each n
  - After training, we can sample from p_θ in reverse: $z_N \rightsquigarrow z_{N-1} \rightsquigarrow \cdots \rightsquigarrow x$
- $p_\theta(z_{n-1} \mid z_n)$ is Gaussian, mean $\mu_\theta(z_{n-1}, n)$ computed by a neural network

Noising model




Noising model
Denoising model
z_9
z_8
z_7
z_6
z_5
z_4
z_3
z_2
z_1
$z_0 = x$
z_9
z_8
z_7
z_6
z_5
z_4
z_3
z_2
z_1
z_0



Generating cats with Gaussian $p_\theta(z_{n-1} \mid z_n)$ requires **hundreds** of steps

Diffusion models: Connections

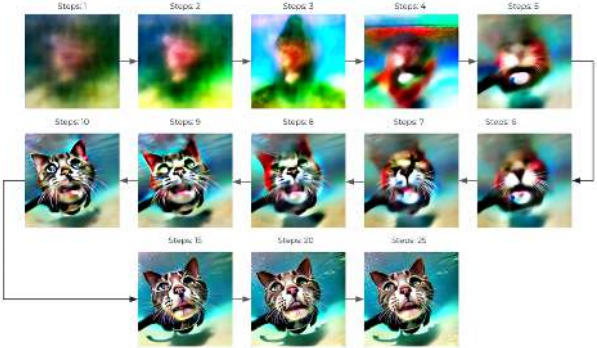
- A diffusion model secretly a VAE whose encoder (q) is fixed

Diffusion models: Connections

- A diffusion model secretly a VAE whose encoder (q) is fixed
- A diffusion model is secretly a continuous-time object

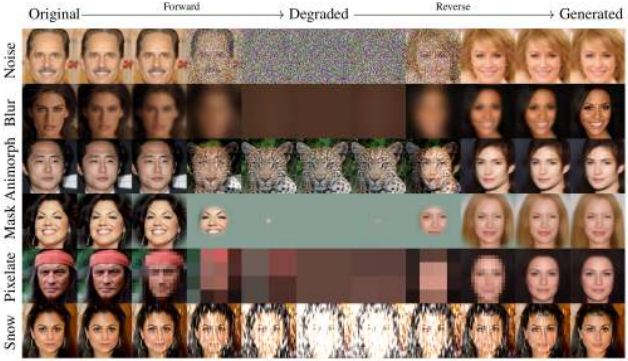
Diffusion models: Connections

- A diffusion model secretly a VAE whose encoder (q) is fixed
- A diffusion model is secretly a continuous-time object
  - Adaptive test-time compute: sample with more steps for better quality



Diffusion models: Generalisations

Diffusion models are much more general than reversing Gaussian noise:



[Bansal et al., 'Cold diffusion. . . ', NeurIPS'23]

We can use any stochastic noising process (but some algebraic simplifications may be lost)

- Revisiting autoregressive language models
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Diffusion for text I: Embedding spaces

Gaussian diffusion in an embedding space

$$\begin{tikzcd} z_N \arrow[r, "p"] & z_{N-1} \arrow[l, dashed, bend left, "q"] \arrow[r, "p"] & \cdots \end{tikzcd} \qquad \begin{tikzcd} z_1 \arrow[r, "p"] & z_0 \arrow[l, dashed, bend left, "q"] \arrow[r, "p(x|z_0)"] & x \arrow[l, dashed, bend left, "q(z_0|x)"] \end{tikzcd}$$

Diffusion for text I: Embedding spaces

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learned decoding

(discrete)

learned embedding

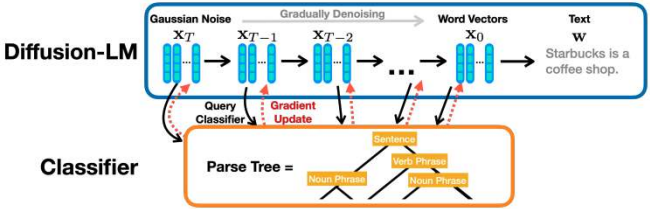
Diffusion for text I: Embedding spaces

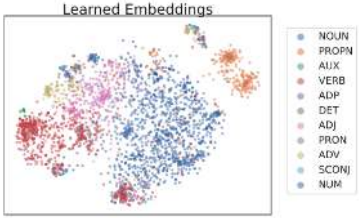
Gaussian diffusion in an embedding space

$$z_N \underset{q}{\overset{p}{\rightleftarrows}} z_{N-1} \underset{q}{\overset{p}{\rightleftarrows}} \cdots \qquad z_1 \underset{q}{\overset{p}{\rightleftarrows}} z_0 \underset{q(z_0|x)}{\overset{p(x|z_0)}{\rightleftarrows}} x \text{ (discrete)}$$

learned decoding

learned embedding





[Li et al., 'Diffusion-LM...', NeurIPS'22]

Diffusion for text II: Discrete-space diffusion

Replace Gaussian noise by random transitions (token replacement or masking)

$$z_N \xrightarrow{p} z_{N-1} \xrightarrow{p} \cdots \qquad z_1 \xrightarrow{p} z_0$$

$$\overline{q} \qquad\qquad \overline{q}$$

discrete transition discrete transition

Diffusion for text II: Discrete-space diffusion

Replace Gaussian noise by random transitions (token replacement or masking)

$$\begin{tikzcd} z_N \arrow[r, "p"] & z_{N-1} \arrow[l, bend left, "q"] \arrow[r, "p"] & \cdots \end{tikzcd} \qquad \begin{tikzcd} z_1 \arrow[r, "p"] & z_0 \arrow[l, bend left, "q"] \end{tikzcd}$$

discrete transition discrete transition

What noising process q to use?

- Our unmasking model corresponds to a q that masks one token at a time

The cats are asleep	$\rightsquigarrow$ The cats $\boxed{\text{M}}$ asleep	$q(z_1 \mid z_0)$
	$\rightsquigarrow$ $\boxed{\text{M}}$ cats $\boxed{\text{M}}$ asleep	$q(z_2 \mid z_1)$
	$\rightsquigarrow$ $\boxed{\text{M}}$ $\boxed{\text{M}}$ $\boxed{\text{M}}$ asleep	$q(z_3 \mid z_2)$
	$\rightsquigarrow$ $\boxed{\text{M}}$ $\boxed{\text{M}}$ $\boxed{\text{M}}$ $\boxed{\text{M}}$	$q(z_4 \mid z_3)$

Diffusion for text II: Discrete-space diffusion

Replace Gaussian noise by random transitions (token replacement or masking)

$$z_N \underset{q}{\overset{p}{\rightleftarrows}} z_{N-1} \xrightarrow{p} \cdots \qquad z_1 \underset{q}{\overset{p}{\rightleftarrows}} z_0$$

discrete transition discrete transition

What noising process q to use?

- Our unmasking model corresponds to a q that masks one token at a time
- An autoregressive model is also a special case (How?)

Diffusion for text II: Discrete-space diffusion

Replace Gaussian noise by random transitions (token replacement or masking)

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	$\rightsquigarrow$ The $\boxed{\text{M}}$ $\boxed{\text{M}}$ $\boxed{\text{M}}$	$q(z_3 \mid z_2)$
	$\rightsquigarrow$ $\boxed{\text{M}}$ $\boxed{\text{M}}$ $\boxed{\text{M}}$ $\boxed{\text{M}}$	$q(z_4 \mid z_3)$

Diffusion for text II: Discrete-space diffusion

Replace Gaussian noise by random transitions (token replacement or masking)

What noising process q to use?

- Our unmasking model corresponds to a q that masks one token at a time
- An autoregressive model is also a special case (How?)
- We can also mask multiple tokens (set probability of masking each token independently)
  - Allows adaptive sampling time (more steps $\rightsquigarrow$ more dependence among tokens)
  - Fewer model passes than sequence length! (But no attention reuse.)

Diffusion for text II: Discrete-space diffusion

127	[M][M][M][

Diffusion for text II: Discrete-space diffusion

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z_2	M	cats	M	asleep
top $p(w_i \mid z_2)$	The		are	
	My		fell	
	Some		fall	
	Are		not	

z_3	M	M	are	M
top $p(w_i \mid z_3)$	The	many		there
	How	much		you
	Some	old		they
	My	cats		big

Diffusion for text II: Discrete-space diffusion

Replace Gaussian noise by random transitions (token replacement or masking)

What noising process q to use?

- Our unmasking model corresponds to a q that masks one token at a time
- An autoregressive model is also a special case (How?)
- We can also mask multiple tokens (set probability of masking each token independently)
  - Allows adaptive sampling time (more steps $\rightsquigarrow$ more dependence among tokens)
  - Fewer model passes than sequence length! (But no attention reuse.)
- We can also replace by other tokens (*e.g.*, embedding-space neighbours) instead of $\boxed{\text{M}}$

Diffusion for text II: Discrete-space diffusion

Key background references on discrete diffusion:

- [Uria et al., 'A deep and tractable density estimator', ICML'14]
- [Austin et al., 'Structured denoising diffusion. . . ', NeurIPS'21]
- [Hoogeboom et al., 'Autoregressive diffusion models', ICLR'22]
- [Lou et al., 'Discrete diffusion modeling. . . ', ICML'24]

Diffusion for text III: Simplices

A middle way: hidden states are full distributions over tokens

$$\begin{tikzcd} z_N \arrow[r, "p"] & z_{N-1} \arrow[r, "p"] \arrow[l, dashed, bend left, "q"] & \cdots \end{tikzcd} \qquad \begin{tikzcd} z_1 \arrow[r, "p"] & z_0 \arrow[l, dashed, bend left, "q"] \end{tikzcd} \ (\approx\text{one-hot})$$

noise logits noise logits

Diffusion for text III: Simplices

A middle way: hidden states are full distributions over tokens

$$z_N \xrightarrow{p} z_{N-1} \xrightarrow{p} \cdots \qquad z_1 \xrightarrow{p} z_0 \ (\approx\text{one-hot})$$

$$\xleftarrow{q} \qquad \xleftarrow{q}$$

noise logits

t	$\operatorname{argmax} \boldsymbol{w}^{c:c+B}_{\text{logits},t}$	$\operatorname{argmax} \hat{\boldsymbol{w}}^{c:c+B}_{t-1}$
2500	of the to the the the the the the the the the the the the the the the the the the the the	apeshifteriao41 fleeting frontman Nutdrop278temp Drama lime Employee cuc rival greatest kan snakes431 cav dreamedRange alloy originally Pact
1500	is the to be the, of the,,,\n the the the the the the the the the\n into the.	stunnedchildrenmetrywaveopensLayer Porn woman transcend242 Homs PluginNext Endsackle microbi spokesperson Brunswick awards":- Sharma Pinball Jr Rug wrapped
1300	of the mission it the as a, for the,,, as to as the the moons, and Cass Cassini is	178 whit promoters du basketballiche SchoolsPur Sack reward basketball corn////WeaponSpeaking squid Chains Caucasian McGivity Me SC rafthr jihadist
1100	was based on the in, 2014. Theini will be the the up is the the the the, Hubble but the the	battles swore starters test thanpadding ambiguityFri BADuitous Stuff depiction bankrupt >>> conversions240Genelvet aptLegweight Riy modesitanesday
900	of the Jarminiini Cass Gr, was supposed to be the most ambitious and most attempt to capture all most distant moons	Sim bag Ves serotonin._ Fab gameplay ransom Alisonorks Fargo expand Rhode pursuing most plagued formulateheter plainly troubled Professional Binary Creek geared
800	is all about Saturn. The Eini will, the closest that the instruments have reached will be to stop in on the Saturn	omial allcounter Saturn. The Directthank Ecuador two thelearning that the Animation have brothers will make toousands downtown governance the Further
700	will allow the Cass to finally see the planet's relatively small atmosphere and finally be able to procure an accurate way of understanding how	willPocket prelim Klux to finally see the planet intelligent relatively jumper atmosphere and halted Fly activityvirt00000 trem accurate way of Inferno what
600	will allow the scientists to better study the effects of Grand Impact, and also be able to get much more data and images of	will allowert scientists Damien better study the effects of Grand Impact, andasket bebery to get much more data and images of
500	will allow the scientists to better see the interior of its atmosphere, and also be able to get much more knowledge and understanding of	will allow the scientists to better see the interior of its atmosphere, and also be able to get much more knowledge and understanding of
1	will allow the scientists to better see the interior of its atmosphere, and also be able to get much more knowledge and observations of	will allow the scientists to better see the interior of its atmosphere, and also be able to get much more knowledge and observations of

[Han et al., 'SSD-LM...', ACL'23]; *see also* [Han et al., 'David helps Goliath...', NAACL'24]

- Revisiting autoregressive language models
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Some practical considerations

- Which kind of diffusion model to use (continuous or discrete)?

Some practical considerations

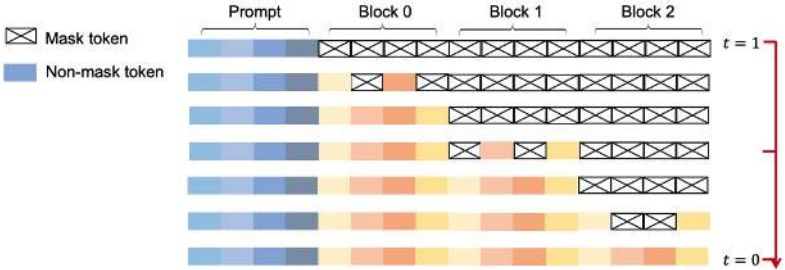
- Which kind of diffusion model to use (continuous or discrete)?
  - Continuous (embedding space and simplex): better understood, but architectures developed for discrete data seem less effective
  - Discrete: more natural for text, connection to MLMs and autoregressive models, better for RL (see below); possibly slower, harder to train and distill

Some practical considerations

- Which kind of diffusion model to use (continuous or discrete)?
- How to handle variable-length text?

Some practical considerations

- Which kind of diffusion model to use (continuous or discrete)?
- How to handle variable-length text?
  - Block-autoregressive diffusion: diffusion model generates blocks of fixed length conditioned on the previous blocks



[Nie et al., 'Large diffusion language models', χ:2502.09992]

Some practical considerations

- Which kind of diffusion model to use (continuous or discrete)?
- How to handle variable-length text?
  - Block-autoregressive diffusion: diffusion model generates blocks of fixed length conditioned on the previous blocks
- Can we do constrained generation and RLHF with diffusion models?
  - Yes: it's a little more complicated mathematically, but may be worth it

A closer look at some big diffusion models

Plaid [Gulrajani & Hashimoto, 'Likelihood-based diffusion. . . ', NeurIPS'23]

- Public 1B-parameter model trained on on OpenWebText
- Embedding-space diffusion

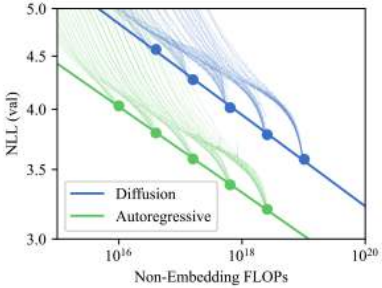


Figure 2: Plaid models improve with compute at a similar rate to autoregressive models, but Plaid is less efficient by a constant factor of 64×.

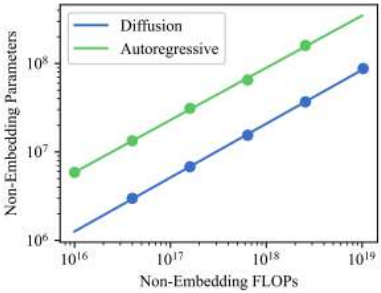


Figure 3: Compute-optimal Plaid models should be 4× smaller (and trained for 4× longer) than compute-optimal autoregressive models.

A closer look at some big diffusion models

Plaid [Gulrajani & Hashimoto, 'Likelihood-based diffusion. . . ', NeurIPS'23]

	PTB (PPL)	enwik8 (BPC)	text8 (BPC)	WikiText2 (PPL)	WikiText103 (PPL)	1BW (PPL)
Plaid 1B (ours)	74.33	1.18	1.12	29.42	28.28	77.64
GPT-2 124M	87.97	1.24	1.22	35.01	35.92	87.85
GPT-2 345M	64.92	1.09	1.11	26.80	26.13	67.34
GPT-2 762M	53.42	1.04	1.06	23.30	22.24	59.48
GPT-2 1.5B	47.59	1.00	1.02	21.33	20.13	54.09

Table 3: Chosen unconditional samples from Plaid 1B demonstrate fluent syntax and long-range coherence. See Appendix D for un-picked random samples.

New research rolled out at an annual scientist meeting finds that the industry will need to recover between 4,000 and 10,000 tons every year of fracked and produced oil and gas fossil reserves in order to do that, according to an analysis done by lead author Dr. Ernesto Monteiro of the University of Alberta in Canada.\n\nThe eye-watering figure represents the total amount of oil and gas - shale and natural gas produced, extracted and sold - will likely need to be recovered in coming years to meet the carbon mitigation goals. ...[698 words omitted]... the team concluded in a report published in an academic journal prepared for the annual meeting of the National Academies of Sciences.

The Barcelona Golf Course doesn't look like a golf course, but it is an oasis for gardening in this busy city.\n\nIt's a massive course spread over 120 acres with fairways that are split right down the middle, unlike the designs on most golf courses. The course enjoys the stunning view of the skyline above it.\n\nA giant oak tree almost 40 meters in diameter serves as one main highlight to the golf course's design. ...[414 words omitted]... The new golf course is accessible on the area's busy streets with shops and restaurants, so the community can enjoy all the leisure activities in the green space.\n\nThe team uprooted the previously existing Pérez Tree, to make room for the new trees to complement Gillet Park.

A closer look at some big diffusion models

Plaid [Gulrajani & Hashimoto, 'Likelihood-based diffusion. . . ', NeurIPS'23]

Prefix completion:

Generative models of text are very versatile: they can be used as a data classification model and also incorporated into multiple data processing engines.\n\nIn this article, we present two new neural memory models capable of processing terabytes of data and the neural networks and computational techniques that are used in those models.

Infilling:

A year ago in Paris, prior to the tournament, I went to Elijah's to eat and get drunk. Everyone in the venue was seventeen. I was there for a few minutes and then I went back to the event. Wow, what a great day! So relaxed and too happy. I do not think I was always like that.

Token-level weights (5× weight on "law"):

Let's talk about law and medicine.\n\n\nIn her dissent, Justice Ron Sen, a veteran administrative law judge, points out that the decision "ignores the fact that the original separation agreement was reached by binding arbitration" that responded to "the legitimate ethical concerns of the university administration," which is what lies "at the heart of law and medicine."

Token-level weights (5× weight on "medicine"):

Let's talk about law and medicine.\n\nIn part because of advancements in technology, personal information about medical and drug use is spreading. Healthcare professionals across the nation rely on this personal data to make decisions about drug prescriptions and clinical trials and monitor people at immediate risk of serious or chronic diseases.

Lexical constraints ("Donald" anywhere):

Also facing legal challenges is Donald Trump's executive order banning immigration from seven Muslim-majority countries that is facing a temporary halt, with nothing scheduled to go into effect. Two federal judges have ruled that such an order violates the establishment clause.

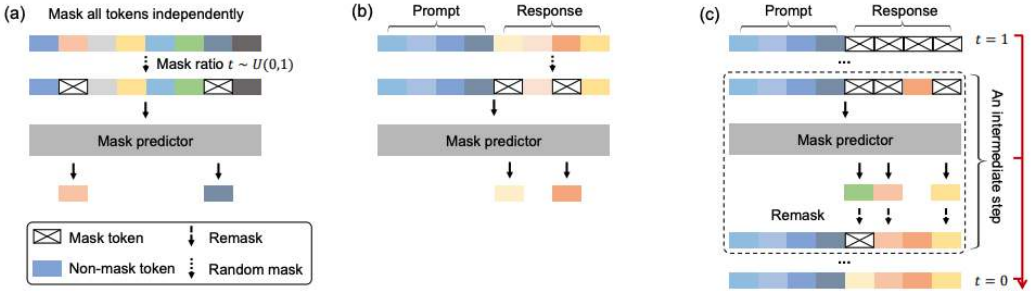
Composition and negation ("Donald" anywhere and "Trump" nowhere):

A month later, with little time to spare, the government hired Donald V. Davis, a former senior aide to Senator Tom Mondale of Minnesota and former Chief Security Operations Officer at the White House, to lead tactical centers.

A closer look at some big diffusion models

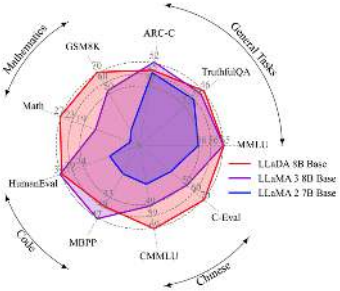
LlaDa [Nie et al., 'Large diffusion language models', χ:2502.09992]

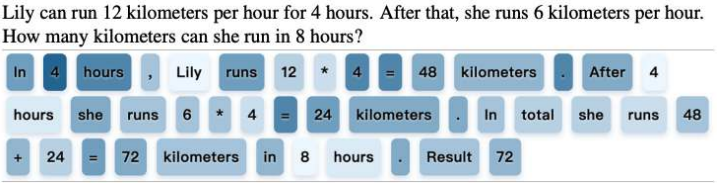
- Public 1B- and 8B-parameter Llama-like models
- Discrete-space diffusion
- Variant fine-tuned for instruction-prompt-response tasks



A closer look at some big diffusion models

LlaDa [Nie et al., 'Large diffusion language models', χ:2502.09992]





A closer look at some big diffusion models

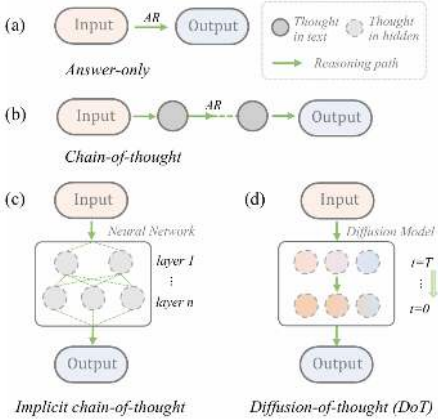
LlaDa [Nie et al., 'Large diffusion language models', χ:2502.09992]

User	Could you please share with me what the initial two lines of the renowned poem 'The Road Not Taken'?
LLaDA	Certainly! The initial two lines of the renowned poem "The Road Not Taken" by Robert Frost are: "Two roads diverged in a yellow wood, And sorry I could not travel both." These lines set the stage for the poem, introducing the idea of a speaker facing a choice between two paths.
User	Please help me translate into Chinese.
LLaDA	Sure! The first two lines of "The Road Not Taken" by Robert Frost can be translated into Chinese as: "两条路分岔在黄色的树林中，遗憾我不能同时走"
User	Please translate into German.
LLaDA	Sure! The first two lines of "The Road Not Taken" by Robert Frost can be translated into German as: "Zwei Wege trennten sich im gelben Wald, und leider konnte ich nicht beide tun."
User	Please also write a poem about life choices. I require 5 sentences, and the first word of each sentence starts with C as the first letter.
LLaDA	Certainly! Here's a poem about life choices, with each sentence starting with the letter "C": Choices cascade through life's journey, Creating connections with each decision, Careful consideration leads to better paths, Courageous choices shape our destiny, Choosing wisely guides us through the years.

Beyond unconditional text generation

Diffusion models can be used for reasoning and planning:

- Latent reasoning chains



When Freda cooks canned tomatoes into sauce, they lose half their volume. Each 16 ounce can of tomatoes that she uses contains three tomatoes. Freda's last batch of tomato sauce made 32 ounces of sauce. How many tomatoes did Freda use?

<<32*2=64>> <<64/16=4>> <<3*4=12>> #### 12

<<32/16=2>> **<<12/12=2*3 ## *2=2= ######**

<<32/16=2>> **#2*3=4>> #4*3=3>>>>>>#######**

<<32/16=2>> **<<2*3=6>> <<6*3=24>> #### 18**

<<32/16=2>> **<<2*3=4>>** <<4*3=12>> #### 12

<<32/16=2>> <<2*2=4>> <<4*3=12>> #### 12

:

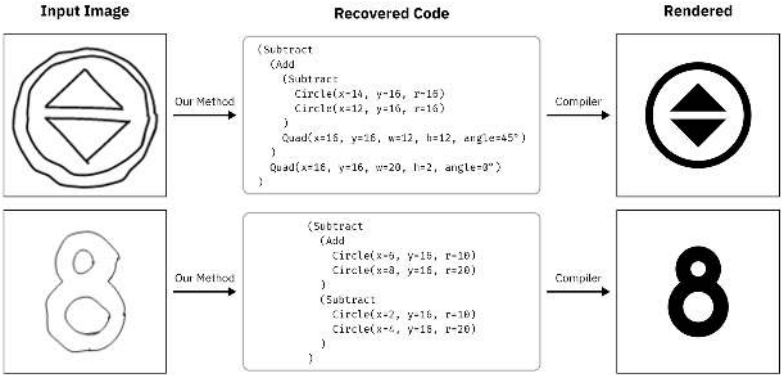
<<32/16=2>> <<2*2=4>> <<4*3=12>> #### 12

[Ye et al., 'Diffusion of thought...', NeurIPS'24]

Beyond unconditional text generation

Diffusion models can be used for reasoning and planning:

- Latent reasoning chains
- Programs (as strings or syntax trees!)



[Kapur et al., 'Diffusion on syntax trees. . . ', χ:2405.20519]

Beyond unconditional text generation

Diffusion models can be used for reasoning and planning:

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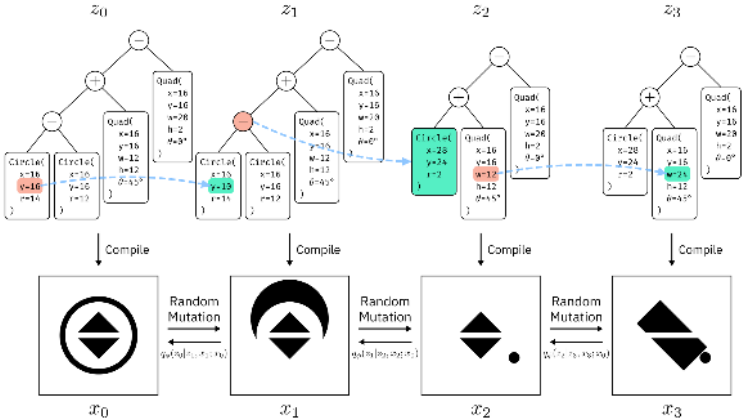

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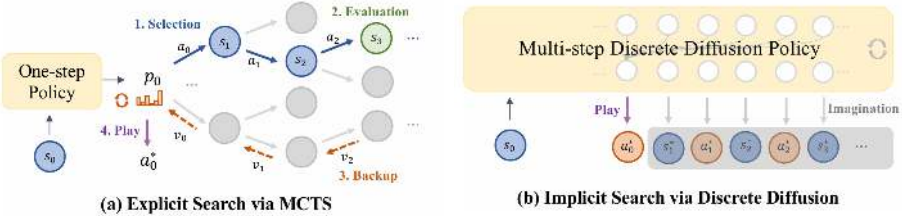

[Ye et al., 'Implicit Search via Discrete Diffusion', ICLR'25]

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- Revisiting autoregressive language models
  - A latent variable view
  - Generation order and masked language modelling
- Very brief introduction to diffusion models
- Three approaches to diffusion for language
- Practical considerations and the state of the art
  - Beyond unconditional text generation
- Remarks and conclusion

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Model	LAMBADA	WikiText2	PTB	WikiText103	1BW
GPT-2	**45.04**	42.43	138.43	41.60	**75.20**
SEDD Absorb	$\leq$50.92	$\leq$**41.84**	$\leq$**114.24**	$\leq$**40.62**	$\leq$79.29
SEDD Uniform	$\leq$65.40	$\leq$50.27	$\leq$140.12	$\leq$49.60	$\leq$101.37
D3PM	$\leq$93.47	$\leq$77.28	$\leq$200.82	$\leq$75.16	$\leq$138.92
PLAID	$\leq$57.28	$\leq$51.80	$\leq$142.60	$\leq$50.86	$\leq$91.12

It's not that long. Lamb stares despairingly at at least two people, amid pale-aged woodland and piles of campsites he now uses as an Atlanta Herald-Western reporter punching out his weary eyes.
"When many of these days went dark none of these forms could go forward without the offenders being high."
"At a few weeks my doctor came at home and had a camera and a book on the marijuana I was taking a young child," says Lamb, now named Sharon Schlessy. She believed that her mother's shaky health had gone on for a moment, but she couldn't do anything about it, but her stepmother lay dead in front of her. But nothing settled with some of her victims. Her mother shot her "every day with a bullet." Three weeks later, Lamb's came back again. "You cut down folks on trees," says the woman with her hair. "Every gun to drow on fruit trees right there was cheap, illegal, and on your own."

[Lou et al., 'Discrete diffusion modeling...', ICML'24] – *example chosen for illustration*

Are diffusion models really better?

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  - $\rightsquigarrow$ much ongoing work on improved architectures and generative processes, distillation, ...
- Often get higher likelihood, but unnatural samples (Why?)
  - Perplexity does not capture all desired properties of text (except in the limit of infinite data)
  - Inductive biases of autoregressive models are closer to human language production and comprehension

Conclusion and outlook

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Questions?